

Subject Description Form

Subject Code	ME42012
Subject Title	Advanced Robotics Studio
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: EIE3320 Object-Oriented Design and Programming
Objectives	To acquire knowledge and hands-on experience of the latest technologies used in robotics. To understand the design process of robots and the systematic approach to solve engineering problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - systematically analyse a given engineering problem and devise an effective robotic solution to address it. - understand and utilise the latest computational technologies to command the motion of robots. - develop computer-aided designs and simulations of robotic systems, mechanisms, and environments. - develop a working robotic prototype by integrating sensors, actuators, mechanisms, and computer systems. - effectively work in teams, design systems, conduct experiments, and deliver effective solutions.
Subject Synopsis/ Indicative Syllabus	Robot Design Process. Robotics as a scientific discipline. Discuss different cases of study and the need for robots in society. Introduce the theory and principles for a methodological robot design. Problem formulation, requirements, and constraints. Computational Tools. Mathematical and numerical preliminaries. Introduce different computer libraries used in robotics. Computer-aided simulation of robotic systems and mechanisms. Digital twins and physics engines. Implementation of common algorithms for perception and control. Hardware Systems. Study the fundamental differences between a computational models and real robotic mechanisms. Study common hardware components (materials, mechanisms, sensors, electronics) used to develop real robotic platforms. Implement Sim2Real strategies to command the motion of real robotic platforms. Course Projects. Conduct hands-on experimental work where students can apply different concepts studied during the course. Form teams and design innovative mechanisms and automated systems that address a particular challenge provided in class. Organise a final course teams' competition where each team presents their unique solution. Field Trip. Group visit to a company, research centre, or institution relevant to the development of advanced robotic systems.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required to design and develop efficient robotics solutions to real-world problems. (Outcomes a to d) 2. Tutorials aim at enhancing students’ technical skills in robotic systems. Students will be able to use the latest robotics technologies with the knowledge acquired in class. (Outcomes b to d) 3. The experiments/projects aim to provide hands-on experience for developing robotic systems and reinforcing the acquired knowledge. (Outcomes b to e) <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="5">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Tutorials</td><td></td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Project or experiments</td><td></td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes					a	b	c	d	e	1. Lectures	√	√	√	√		2. Tutorials		√	√	√		3. Project or experiments		√	√	√	√											
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Reading List and References	• Lecture notes, articles, and handouts. • Practical and Experimental Robotics. F. Sahin and P. Kachroo, CRC Press, Latest Edition. • Engineering Design, by Rudolph J. Eggert. High Peak Press, Latest Edition. • Practical Field Robotics: A Systems Approach, by Robert H. Sturges. Wiley Press, Latest Edition. • Programming Robots with ROS: A Practical Introduction to the Robot Operating System, by Morgan Quigley. O'Reilly Media, Latest Edition. • Arduino Programming Notebook. B. Evans, Latest Edition.
Develop Date	June 2024